

Certified Non-Sharpness of a Hankel Bound for $\pi/3$ -Spirallike Convex Functions

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Abstract

Let $\mathcal{C}_{\pi/3}$ be the normalized, locally univalent functions on the unit disk satisfying

$$\operatorname{Re}\{e^{i\pi/3}(1 + zf''(z)/f'(z))\} \geq 0,$$

and let $M_{\pi/3} = \sup_{f \in \mathcal{C}_{\pi/3}} |a_2a_4 - a_3^2|$. We prove the certified strict enclosure

$$\frac{9291}{320000} < M_{\pi/3} < \frac{4181}{144000}.$$

In particular, the previously available estimate 89/2880 is not sharp at $\pi/3$. The proof gives an exact Schur-parameter reduction, uses an explicit feasible rational witness for the lower bound, and supplies a finite exact-rational interval certificate for the upper bound. The exact value of $M_{\pi/3}$ and the classification of its extremizers remain unresolved.

1 Result and context

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. The class considered here consists of functions

$$f(z) = z + \sum_{n \geq 2} a_n z^n$$

that are holomorphic and locally univalent in $\mathbb{D}$ and satisfy

$$\operatorname{Re}\left[e^{i\pi/3}\left(1 + \frac{zf''(z)}{f'(z)}\right)\right] \geq 0 \quad (z \in \mathbb{D}). \quad (1)$$

The second Hankel functional is

$$H_{2,2}(f) = a_2a_4 - a_3^2, \quad M_{\pi/3} = \sup_{f \in \mathcal{C}_{\pi/3}} |H_{2,2}(f)|.$$

Krishna and Reddy obtained a parameter-dependent estimate for this functional on spirallike convex classes [2, Theorem 3.3]. In the notation above its specialization is

$$M_{\pi/3} \leq \frac{17(1 + \cos^2(\pi/3)) + 2\cos(\pi/3)}{144(1 + \sec^2(\pi/3))} = \frac{89}{2880}. \quad (2)$$

Ahamed, Sarkar, and Das restated this one-variable estimate while studying a Banach-space extension, proved sharpness at parameter zero, and explicitly left the exact nonzero-parameter variational problem unresolved [1].

Our contribution has deliberately narrower scope than a solution of that variational problem. At the level of methodology only, angle-dependent complex reduction followed by exact rational certification also appears in the distinct convex-lens spectral problem studied by Zeng [3]. That work concerns operator spectral-set constants and supplies no bound for the Hankel functional considered here.

Theorem 1. *For the class $\mathcal{C}_{\pi/3}$ defined by (1),*

$$\frac{9291}{320000} < M_{\pi/3} < \frac{4181}{144000}. \quad (3)$$

Consequently, the upper estimate (2) is strictly non-sharp at $\pi/3$.

The certified interval in (3) has width $1/2880000$. No assertion of exhaustive priority is intended: the comparison made here is with the two directly inspected sources above.

2 Exact Schur reduction

Let $\mathcal{P}$ denote the Carathéodory class of holomorphic functions

$$p(z) = 1 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots, \quad \operatorname{Re} p(z) > 0 \quad (z \in \mathbb{D}).$$

Set

$$\gamma = e^{-i\pi/3} \cos(\pi/3) = \frac{1 - i\sqrt{3}}{4}.$$

The real part in (1) is harmonic, nonnegative, and equal to $1/2$ at the origin; the strong minimum principle therefore makes it positive throughout $\mathbb{D}$. Condition (1) is equivalent to the existence of a unique $p \in \mathcal{P}$ such that

$$1 + \frac{zf''(z)}{f'(z)} = 1 + \gamma(p(z) - 1). \quad (4)$$

Conversely, every $p \in \mathcal{P}$ gives a member of $\mathcal{C}_{\pi/3}$ through

$$f'(z) = \exp\left(\gamma \int_0^z \frac{p(\zeta) - 1}{\zeta} d\zeta\right), \quad f(z) = \int_0^z f'(t) dt. \quad (5)$$

In particular, local univalence follows directly from the exponential formula.

Coefficient comparison in (4) gives

$$\begin{aligned} a_2 &= \frac{\gamma c_1}{2}, \\ a_3 &= \frac{\gamma c_2 + \gamma^2 c_1^2}{6}, \\ a_4 &= \frac{2\gamma c_3 + 3\gamma^2 c_1 c_2 + \gamma^3 c_1^3}{24}. \end{aligned}$$

Thus

$$H_{2,2}(f) = \frac{\gamma^2}{144} Q, \quad Q = 6c_1 c_3 - 4c_2^2 + \gamma c_1^2 c_2 - \gamma^2 c_1^4. \quad (6)$$

Since $|\gamma| = 1/2$, one has $576|H_{2,2}(f)| = |Q|$.

Under the rotation $p(z) \mapsto p(e^{i\theta}z)$, every term in Q is multiplied by $e^{4i\theta}$. We may therefore put $c_1 = r \in [0, 2]$. With $D = 4 - r^2$, two steps of the Schur algorithm give the closed-disk parametrization

$$2c_2 = r^2 + Dx, \quad 4c_3 = r^3 + 2rDx - rDx^2 + 2D(1 - |x|^2)w, \quad |x|, |w| \leq 1. \quad (7)$$

For completeness, write $p(z) = (1 + zh(z))/(1 - zh(z))$, where h is a Schur function and $h(0) = \alpha = r/2$. Then

$$h(z) = \frac{\alpha + z\psi(z)}{1 + \alpha z\psi(z)}, \quad \psi(0) = x,$$

and Schwarz–Pick gives $\psi'(0) = (1 - |x|^2)w$ for some $|w| \leq 1$. Expansion through degree three gives (7); conversely, the same Schur steps realize every displayed choice. This includes the degenerate boundary cases. Substitution into (6) yields

$$Q = A + 3rD(1 - |x|^2)w, \quad A = \frac{3r^4}{4} + \frac{9 - i\sqrt{3}}{8}r^2Dx - \frac{D(r^2 + 8)}{2}x^2. \quad (8)$$

The coefficient of w is nonnegative, so maximization over its disk is exact:

$$576M_{\pi/3} = \max_{\substack{0 \leq r \leq 2 \\ |x| \leq 1}} \{|A| + 3r(4 - r^2)(1 - |x|^2)\}. \quad (9)$$

For later use, put $s = |x|^2$, write $x = \sqrt{s}(X + iY)$ with $X^2 + Y^2 = 1$, and define

$$a = \frac{3r^4}{4}, \quad c = \frac{(4 - r^2)(r^2 + 8)}{2}, \quad K = r^2(4 - r^2).$$

Choosing the sign of Y that maximizes the only sign-dependent term and expanding exactly gives

$$\max_{Y=\pm\sqrt{1-X^2}} |A|^2 = C + \ell X + b\sqrt{1 - X^2} - qX^2, \quad (10)$$

where

$$\begin{aligned} C &= (a + cs)^2 + \frac{21}{16}K^2s, & \ell &= \frac{9}{4}K\sqrt{s}(a - cs), \\ b &= \frac{\sqrt{3}}{4}K\sqrt{s}(a + cs) \geq 0, & q &= 4acs \geq 0. \end{aligned} \quad (11)$$

3 Finite exact-rational upper certificate

We describe the certificate sufficiently explicitly to separate its exact content from numerical exploration. For any rational $t > 0$, set

$$u = \frac{1 - t^2}{1 + t^2}, \quad v = \frac{2t}{1 + t^2} > 0.$$

For $-1 \leq X \leq 1$,

$$(1 - uX)^2 - v^2(1 - X^2) = (X - u)^2, \quad 1 - uX > 0,$$

and hence

$$\sqrt{1 - X^2} \leq \frac{1 - uX}{v}. \quad (12)$$

For $L, q \geq 0$, define

$$m(L, q) = \max_{|X| \leq 1} (LX - qX^2) = \begin{cases} L, & q = 0, \\ L^2/(4q), & q > 0 \text{ and } L \leq 2q, \\ L - q, & q > 0 \text{ and } L > 2q. \end{cases} \quad (13)$$

Combining (10)–(13) gives, for every chosen tangent, an explicit upper bound for the phase maximum.

The accompanying certificate evaluates these bounds on rational boxes $B = [r_-, r_+] \times [s_-, s_+] \subset [0, 2] \times [0, 1]$ by outward interval arithmetic. All endpoints are fractions. If $h = n/d \geq 0$ and $S = 2^{72}$, it computes

$$k = \left\lfloor \sqrt{\lceil nS^2/d \rceil} \right\rfloor$$

and encloses $\sqrt{h}$ by $[k/S, (k+1)/S]$, using the singleton interval when equality is exact. Products use all four endpoint products, and squaring treats intervals crossing zero separately.

On each box, the expressions in (11) are first evaluated as intervals. For each t , the certificate uses the upper endpoint of C , the upper endpoint of b , the largest absolute endpoint of $\ell - bu/v$, and the lower endpoint of q in (13). Monotonicity in these four quantities makes the result an upper bound. Taking the minimum over the following 50 globally valid tangents preserves that property:

$$\left\{ \frac{1}{10}, \frac{1}{2}, 1, 2, 3, 4, 5, 6, 7, 8, 10, 20 \right\} \cup \left\{ \frac{k}{20} : 100 \leq k \leq 140 \right\}. \quad (14)$$

After an outward square root, the upper endpoint of $3r(4 - r^2)(1 - s)$ is added, as required by (9).

The initial 16×8 rational grid covers the closed rectangle $[0, 2] \times [0, 1]$. Any box whose certified upper bound is not strictly less than $4181/250$ is bisected along its wider normalized coordinate. The deterministic run ends with no unresolved box after processing 31,358 boxes; the maximum reported split depth is 28. Hence a finite collection of verified closed boxes covers the entire parameter rectangle and proves

$$576M_{\pi/3} < \frac{4181}{250}. \quad (15)$$

The endpoint faces $r = 0, 2$ and $s = 0, 1$ are part of the same cover. No floating-point comparison is used to accept a box.

4 Exact feasible witness and proof of the theorem

For the lower bound in (9), choose

$$r = \frac{151}{200}, \quad x = \frac{-35 + 12i}{37}; \quad |x| = 1. \quad (16)$$

These parameters correspond to an actual Carathéodory function, rather than only to a formal coefficient triple. Indeed, with $\alpha = r/2$, the disk automorphism

$$\phi(z) = \frac{\alpha + xz}{1 + \alpha xz}$$

is a Schur function, and

$$p(z) = \frac{1 + z\phi(z)}{1 - z\phi(z)} \quad (17)$$

belongs to $\mathcal{P}$. Expanding (17) gives precisely (7) with the parameters in (16); the last Schur term vanishes because $|x| = 1$. Formula (5) then supplies a feasible $f \in \mathcal{C}_{\pi/3}$.

For transparency, a direct exact calculation of (8) at this witness gives

$$|A|^2 = N_0 + N_1\sqrt{3}, \quad (18)$$

where

$$N_0 = \frac{15453135790961885783319393}{5607424000000000000000}, \quad N_1 = \frac{897410256632116046397}{3788800000000000000000} > 0.$$

Since

$$\sqrt{3} > \frac{265}{153} \quad \left(3 - \left(\frac{265}{153} \right)^2 = \frac{2}{23409} > 0 \right),$$

one obtains by rational arithmetic

$$N_0 + N_1 \frac{265}{153} - \left(\frac{83619}{5000} \right)^2 = \frac{454370647151043061941}{95326208000000000000000000} > 0.$$

Thus the feasible witness satisfies

$$|A| > \frac{83619}{5000}. \quad (19)$$

Equations (15) and (19), divided by 576, give

$$\frac{9291}{320000} < M_{\pi/3} < \frac{4181}{144000},$$

which proves the first assertion of Theorem 1. Finally,

$$\frac{89}{2880} - \frac{4181}{144000} = \frac{269}{144000} > 0,$$

so (2) is strictly non-sharp. This completes the proof.

5 Limitations and reproducibility

The certificate establishes only the strict enclosure (3). It does not prove that a maximizing Schur parameter must satisfy $|x| = 1$, does not identify the exact algebraic value of $M_{\pi/3}$, and does not classify the equality fibers. Those are precisely the remaining parts of the original problem.

The ancillary program `certify_bounds.py` uses only Python's standard library. A fresh run with

```
python3 certify_bounds.py
```

recomputes the witness enclosure and the finite branch-and-bound cover. The reported elapsed time and printed decimal approximations are diagnostics and play no deductive role.

Computational and AI assistance disclosure. AI systems assisted in the algebraic reduction, construction and review of the exact-arithmetic certificate, source-record checking, and preparation of this manuscript. Exploratory floating-point optimization was used to locate promising parameters and to choose a convenient finite tangent set, but no floating-point value is used in the proof. The stated theorem rests on the displayed exact derivation, the explicit feasible witness, and the human-inspectable deterministic certificate. The author takes responsibility for the claims and presentation.

References

- [1] M. B. Ahamed, N. Sarkar, and P. Das, *Second Hankel determinant for β -spirallike convex mappings in complex Banach spaces*, arXiv:2608.07563v1 [math.CV] (2026). <https://doi.org/10.48550/arXiv.2608.07563>
- [2] D. Vamshee Krishna and T. Ram Reddy, *Coefficient inequality for certain subclasses of analytic functions associated with Hankel determinant*, Indian J. Pure Appl. Math. **46** (2015), 91–106. <https://doi.org/10.1007/s13226-015-0111-1>
- [3] Z. Zeng, *Fixed convex-lens spectral constants: Möbius reduction, sharp model theorems, and angle-dependent bounds*, Preprints 202608.1272, version 1 (2026). <https://doi.org/10.20944/preprints202608.1272.v1>